Q19. It has several procers

① tokanization

It convert information into token or small piece that easy to understand

ex. I Love NLP

so tokanization converted into each small piece of information

[I] [Love] [NLP]

② stop-word removal

In this step we remove stop-word like and, a, the, an, etc

ex. I Love NLP and DL

so sentence is ⇒ I Love NLP DL

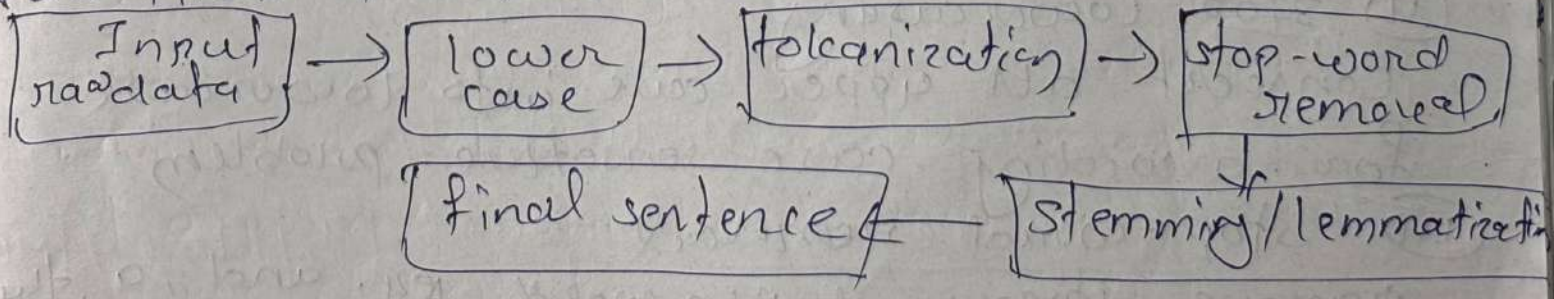
(iii) stemming / lemmatization

In this step we organize all information in proper way so human and machine can understand the actual meaning of sentence

(iv) Lowercasing

This is a step used in Text preprocessing that convert all the uppercase word into lower case to overcome the case sensetive Problem

So final Architecture will be



[redacted] steps in normalization

Normalization is a process in which [redacted] missing word, incorrect sentence and unwanted errors are solve and give a normal or standard sentence so user or machine can understand the data.

* It have almost same process or step as Text preprocessing

① Tokanization
convert raw data into small piece of data

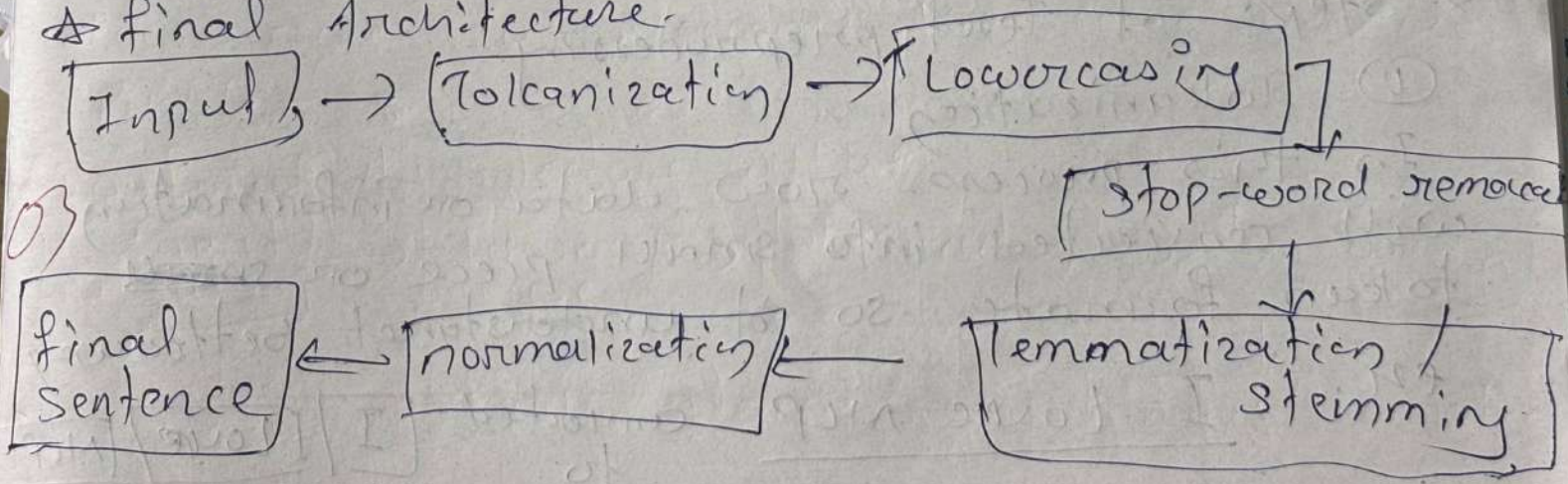
② [redacted] lowercasing
convert all upper case into lowercase for avoiding case sensetive problem

③ stop-word removal
remove stop-word like word is, am, are, and, a, the etc.

(IV) lemmatization / stemming
- adject sentence as it make meaning of sentence

(V) normalizating
evaluate all small token and make a meaningful sentence that machine can be understand.

* final Architecture.



Q.1.b) Text preprocessing

- Text preprocessing is a technique of cleaning and arranging the sentence in human and machine readable form
- In this process we organize unstructured, unordered raw data into structured and ordered way so human and machine can read properly

* steps of Text preprocessing

① Tokanizating

In this process raw data or information will converted into small piece or [redacted] token formate so it understand better way

eg. I Love NLP converted to [I] [Love] [NLP]

(ii) Lowercasing

- In this process a raw data (information) will converted into lowercase
- convert all uppercase letter into lower case for avoiding case sensetive problem

ex. I Love NLP converted to i love nlp

(iii) stop-word removal

- remove word like a, and, the, an, is, am, are. for better understanding and storing purpose

ex. I Love a fruits converted to I Love fruits

(14) stemming / lemmatization

- In this process we organize the information in such away that meaning not change

final Architecture of Text preprocessing

Input
↓
Tokenization
↓
Lowercasing
↓
stop-word removal
↓
stemming / lemmatization
↓
final sentence.

* stages of Text preprocessing

(I) Lexical analysis

(II) Syntactic analysis

(III) semantic analysis

(IV) Discourse Treatment

(V) praymtic analysis

This are the several stages that help to manage or understand the meaning of sentend and not change the meaning of that sentence.

Q. 3. a) categorization of Text

- categorization (also called as classification) is a process in which raw data are organize into various category that help to understand the information and find the information in large data
- Text categorization mince a converting or arranging a large data into category so user can find the information fast
- categorization work on information retrival
- It help to organize document in structured formats

A large information converted to category
ex news documentation converted to various category like sport, Historical, Jeographical, political, and so on
so this are the category e.

- It widly used in organization to organize raw data in proper structure
- classification is a another term that use this raw data for prediction.
It use for spam detection also
- classification classify the informating so user and machine can understand better way.

Q. 3. b) N-gram model

- N-gram models are the mathamatical and statastical models that help to predict next word from privious word
- It predict next word from privious word.

* It categorized into 3 main types

(i) Unigram
(ii) Bigram
(iii) Trigram

(i) Unigram

unigram model predict a word individually It do not take privious word

ex. I love NLP